

Research Statement

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We live in an increasingly interconnected era in which individual entities, such as people, computers, or vehicles, cannot be considered solely in isolation, but must instead be understood as components of *complex systems* whose local and global dynamics are strongly shaped by interactions among their constituent elements. For example, in social networks and, more broadly, within society, individual opinions are influenced by the communities to which people belong [1, 2]. Similarly, in transportation systems, the behavior of a single vehicle may be substantially affected by cascading disruptions caused by accidents or congestion [3, 4, 5].

As a result of these interconnections, control does not necessarily need to be applied independently to every individual component; instead, it can be designed at the network level, exploiting the interactions among components to influence the behavior of the entire system [6, 7]. At the same time, these interactions create vulnerabilities, because faults, corrupted information, or malicious inputs can propagate beyond the components they initially affect [8, 9].

In response to these vulnerabilities, a substantial body of research has emerged over the past fifteen years on the *resilient and secure control of networked systems*. Researchers in this area have addressed several related problems: determining whether a networked dynamical system is under attack and identifying the compromised components [8, 10, 11]; reconstructing the true system state when sensors, communication channels, or measurements have been corrupted [9, 12, 13]; assessing whether trustworthy nodes can still reach agreement or coordinate in the presence of malicious, faulty, or Byzantine agents [14, 15, 16]; designing controllers that preserve stability, safety, and acceptable performance despite attacks, communication failures, or corrupted inputs [17, 18, 11]; and examining how network topology, connectivity, redundancy, and sensor or actuator placement influence system vulnerability [8, 19, 20].

Historically, much of the previously mentioned systems and control research has relied on explicit mathematical models of physical systems. These models may be derived from first principles using known physical laws or estimated from experimental data through system identification. Once such a model is available, control theory provides tools to analyze properties such as stability, controllability, observability, robustness, and optimality, and to design controllers with rigorous performance guarantees.

In practice, however, mathematical models are often imperfect because they cannot fully capture complex dynamics, changing environments, or unknown disturbances. As a result, recent research has increasingly explored hybrid approaches that combine model-based control with learning-based methods [21, 22, 23, 24, 25]. For example, when controlling a drone, Newtonian mechanics can provide an accurate model of its rigid-body dynamics, while a neural network is trained on flight data to learn unmodeled aerodynamic effects [26]. The resulting hybrid model can improve predictive accuracy while retaining the physical structure and interpretability of the nominal model. More broadly, researchers are also incorporating physical knowledge and control-theoretic constraints directly into learning algorithms (such as reinforcement learning) to improve safety, sample efficiency, robustness, and generalization [27].

In other settings, the complexity and uncertainty of the environment make it difficult to construct sufficiently accurate models for decision-making, motivating more direct learning-based approaches, such as autonomous driving in environments populated by unpredictable humans [28, 29], or strategic interactions in cooperative and competitive multi-agent environments [30]. However, the very flexibility that makes learning-based approaches powerful also introduces fundamental challenges: learned models can be difficult to interpret ("*black-box*" models) [31, 32], their behavior may change under distribution shifts [33, 34], and their decisions can be vulnerable to unreliable or adversarial information [35, 36]. These challenges become particularly critical in networked autonomous systems, where individual

agents must rely on information shared by other agents while making decisions that affect the collective behavior of the system.

Building upon these ideas, I am interested in extending the objectives of "classical" resilient control to learning-enabled autonomous agents, where agents must not only operate under adversarial conditions but also learn effective behaviors and interaction structures when system dynamics, communication patterns, and sources of information cannot be fully characterized a priori. More specifically, the fundamental question I would like to try to answer is

How can autonomous agents learn reliable interaction mechanisms and make effective decisions in networked environments when information sources may be unreliable or adversarial?

Fundamentally, these objectives may be *collective*, such as maintaining formation or reaching agreement, *individual but interdependent*, such as each vehicle reaching its own destination safely or each robot completing its assigned task, or *system-level*, such as preserving power delivery or traffic flow.

For example, consider a network of autonomous vehicles navigating a shared environment. Each vehicle must make decisions based not only on its own sensors but also on information communicated by nearby vehicles, such as their trajectories, intentions, or observations of surrounding conditions. While cooperation among vehicles can improve safety and efficiency, a compromised or unreliable vehicle may provide inaccurate information, either due to failures or intentional manipulation. Many classical resilient-control formulations assume that the system or communication structure is known and impose a predefined model or bound on adversarial behavior [8, 9, 14]. Learning-enabled agents must determine from experience which information sources are reliable [37] while simultaneously learning how to accomplish their assigned objectives. This raises fundamental questions about how trust, communication, and decision-making should be learned in adversarial networked environments.

Another interesting research direction aligned with these interests concerns human-robot interaction. My previous work on human behavior modeling [38, 39] helped me appreciate how difficult it can be for autonomous systems to anticipate human actions: humans are not only inherently unpredictable, but may also act strategically or maliciously. For example, consider a robot operating alongside a human who may unexpectedly change their behavior, either because of an unforeseen circumstance or because they intentionally seek to manipulate the robot's decisions. A reliable autonomous system should therefore not only account for uncertainty in human behavior, but also learn to distinguish between benign and potentially adversarial interactions and adapt its decisions accordingly. I am particularly interested in developing learning and control methods that allow robots to reason about the reliability of information and human behavior while maintaining safety and performance under uncertainty. This perspective extends the notion of resilient decision-making beyond failures or compromised network agents to settings in which the most difficult and unpredictable component of the system is the human itself.

Ultimately, I hope to develop autonomous systems that can learn to make reliable decisions not only when interacting with well-characterized environments and agents, but also when operating in the uncertain, dynamic, and sometimes adversarial environments of the real world.

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